

What would have to be true for a 100 MW onshore wind farm in Finland to reach FID?

A 100 MW onshore wind farm with COD 2028, and an optional co-located 25 MW / 50 MWh battery

Take FID only with a ≥ 15 -year corporate PPA at $\geq \sim 70$ EUR/MWh

Merchant onshore wind in Finland does not finance. CAPEX and site quality move value more than the long-term price view.

17 % gearing

Merchant does not finance

Lenders fund only 17 % of CAPEX without a PPA (47 % with one). Best merchant scenario: 2.1 % equity IRR.

70–74 EUR/MWh

The PPA price decides

Break-even 70.35 (NPV = 0) to 73.94 EUR/MWh (10 % equity IRR), vs 29–50 in public PPA indices.

–2.5 EUR m

The battery is a separate bet

Standalone BESS needs 78 kEUR/MW/yr vs 89 today; it does not change the wind economics.

Finnish prices: a high average, a low median and futures near 50 EUR/MWh

Day-ahead prices, FI (EUR/MWh, hourly)

Period	Mean	Median	Neg. hours
2023	56.5	42.8	467
2024	45.6	27.6	725
2025	40.5	16.7	465
Oct 25–Sep 26	55.8	34.6	104

In 2025 the mean was more than twice the median: value sits in a few scarcity hours.

Forward and fundamental views (EUR/MWh)

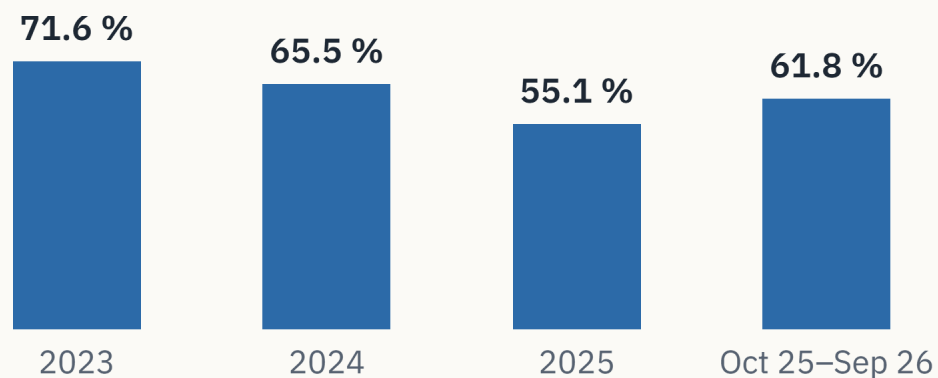
Reference	Value
EEX Cal-28, nominal / real 2026	56.0 / 53.8
EEX Cal-30, nominal / real 2026	52.7 / 48.7
Lampela (2023), Base 2030	39.0
KYOS baseload, 2027–36	60.0
Model base path, 2027–36 (real)	48.2

Model base path: futures to 2030, thesis from 2040. It sits 6–12 EUR/MWh below KYOS, i.e. conservative.

Finnish wind earns only about 60 % of the average price

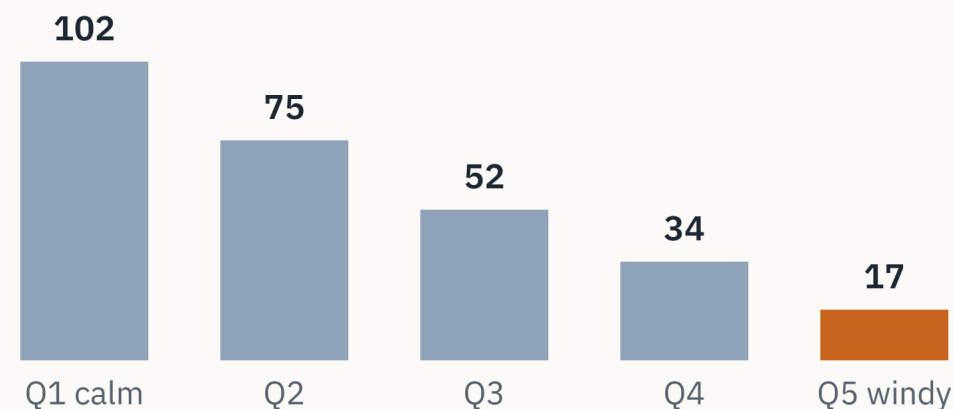
Wind capture rate

Wind capacity in Fingrid data +60 % (5.8 → 9.4 GW)



Price by wind-output quintile

EUR/MWh, Oct 2025–Sep 2026



KYOS outlook 2027–36, implied wind capture rate: **Finland 60 %**, the lowest zone · DE 83 % · SE3 85 % · DK1 88 %

Source: own analysis of ENTSO-E prices and wind (FI, hourly), fleet profile; capture rate with no output at negative prices: 80.6 / 74.5 / 59.1 / 63.0 %. Fingrid dataset 268. KYOS PPA Insights No. 19 (Jun 2026).

INVESTMENT CASE · INPUTS

Every input carries a source and a status

Parameter	Base case	Main source	Status
Capacity	100 MW	Finnish projects 105–455 MW	own assumption
COD / lifetime	2028 / 25 years	Renewables Finland (2023); OX2	source-based
Net capacity factor	35 % (P50)	OX2 38.7 %; NIB 31.7 % (P90)	source-based
Curtailement	5 %	Own analysis, negative-price hours	source-based
CAPEX	1,500 EUR/kW	OX2 FIDs 2025; WindEurope 2026	source-based
OPEX incl. lease	44 EUR/kW/yr	NREL, EWEA, DEA; FI property tax	source-based
PPA volume / tenor	100 % / 15 years	OX2 FIDs; Google EU wind PPAs	source-based
PPA price	50 EUR/MWh, fixed	LevelTen P25 FI+PT: 49.95	own assumption
Debt rate / tenor	5.75 % / 15 years	EUR swap 3.7 % + 2.0 % margin	source-based
Minimum DSCR	1.30x (PPA)	Secondary sources only	own assumption
WACC	6 % PPA / 7 % merchant	CAPM build-up ≈ 5.9 %	own assumption
Corporate tax	20 %	vero.fi	verified

Source: model tab Inputs (full source text per cell). 43 wind inputs: 3 verified, 24 source-based, 16 own assumption, 0 to verify. Prices and costs in real 2026 EUR, indexed at 2 % (ECB). Tax depreciation 23.5 % declining balance (KPMG Law 2023).

BANKABLE OR NOT?

At 50 EUR/MWh the project destroys value; merchant is not bankable

Base case: 100 % PPA at 50 EUR/MWh

-46.5

Project NPV, EUR m (6 %)

1.6 %

Project IRR, nominal

-0.3 %

Equity IRR, nominal

47.9

LCOE, EUR/MWh real 2026

Fixed nominal 50 EUR/MWh $\approx$ 45 EUR/MWh indexed (own calculation), below the LCOE; after 2043 the plant sells merchant.

PPA vs merchant

Metric	100 % PPA	Merchant
DSCR (sizing)	1.30x	1.60x
WACC (nominal)	6 %	7 %
Gearing	47 %	17 %
Equity IRR, Base/Base	-0.3 %	-3.4 %
Equity IRR, 9 scenarios	-5.4 to 2.6 %	-12.1 to 2.1 %
NPV, 9 scenarios (EUR m)	-55 to -33	-131 to -51

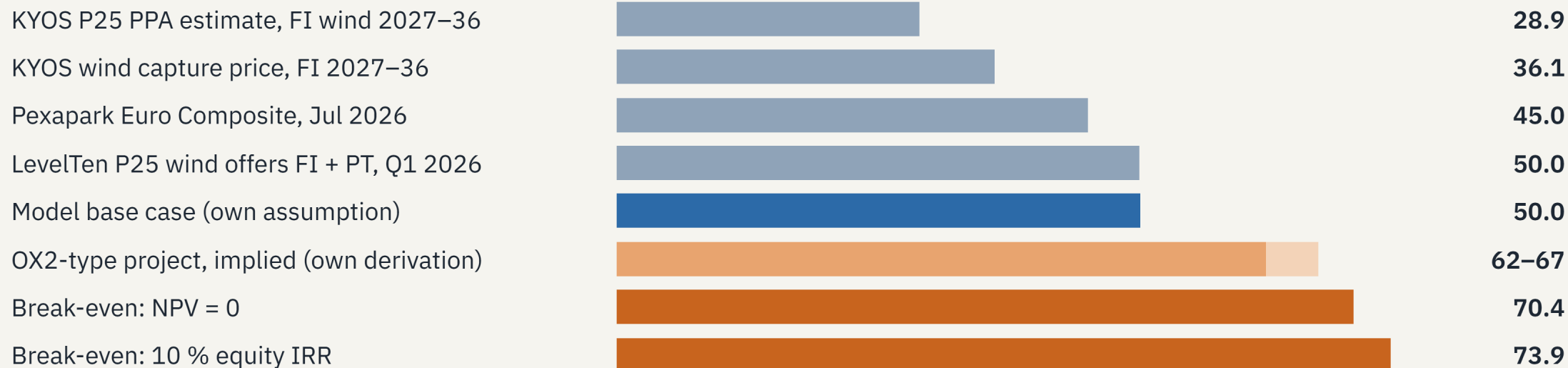
Debt capacity, not cost, is the binding constraint without a PPA.

Source: own model v1.8, Outputs rows 5–26 (scenario grids are snapshots of full model runs). Scenarios = market price level $\times$ capture rate (Low / Base / High). NPVs at 6 % and 7 % are not directly comparable; equity IRRs are.

WHAT WOULD HAVE TO BE TRUE · PPA PRICE

The required PPA price is 70–74 EUR/MWh, above public price signals

EUR/MWh, Finland onshore wind

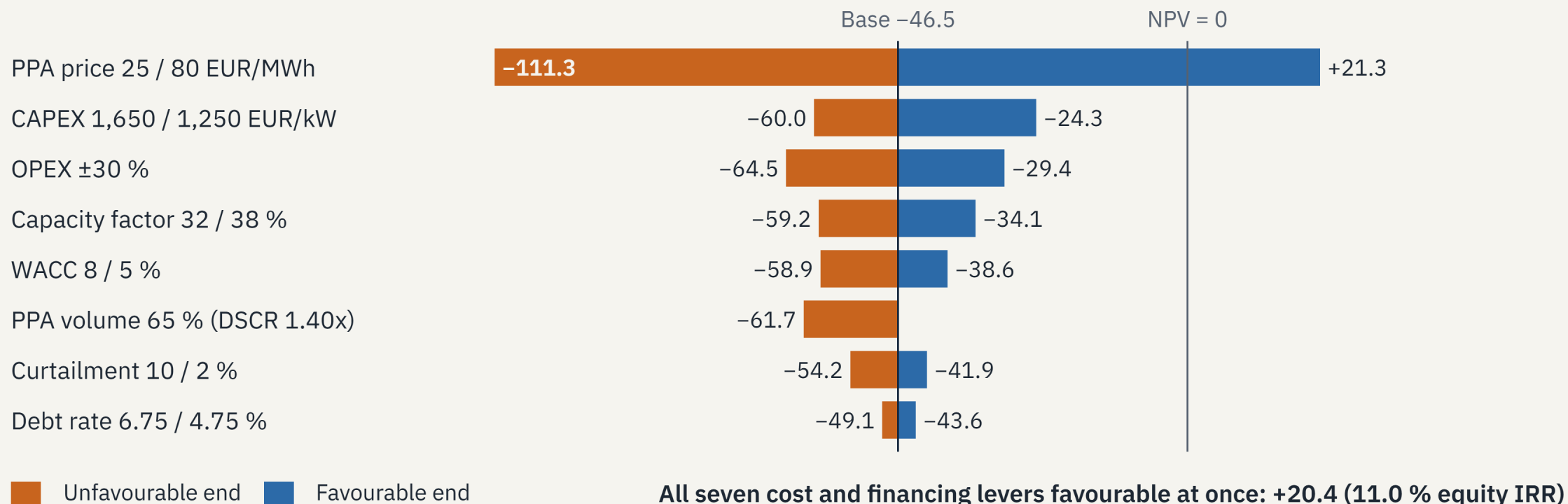


Gap to the base case: 20–24 EUR/MWh. At break-even the project IRR equals the 6 % WACC and gearing reaches 72–75 %.

Source: own model v1.8, Outputs rows 28–32 (100 % PPA, 15 years, fixed nominal, DSCR 1.30x; solved by iteration incl. debt re-sizing). KYOS PPA Insights No. 19 (Jun 2026); Pexapark (Jul 2026, all technologies, Europe); LevelTen via SR Inc (Q1 2026, seller offers; FI wind steady in Q2 2026). OX2 value: own model runs with OX2 parameters, PPA prices not published.

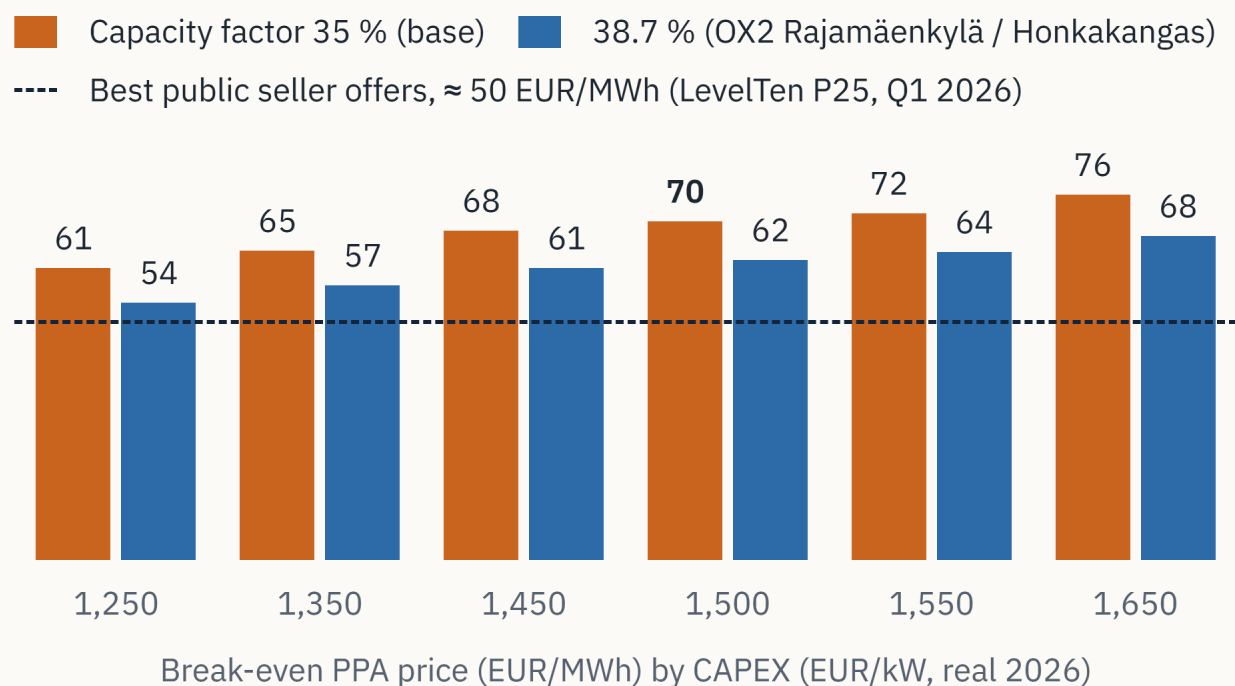
SENSITIVITIES · PROJECT NPV, EUR M

No single cost lever closes the gap; only the PPA price does



Source: own model v1.8, single-lever runs around the base case (bands from Inputs notes), full recalculation, 4 Oct 2026. OPEX band ±30 % is wide. Debt rate acts on NPV via the WACC only (linked, ±0.38 pp). Combined case: own judgement, unrealistic.

Lower CAPEX and a better site cut the required PPA price, but not to market level



Trade-off

–100 EUR/kW CAPEX $\approx$ –3.5 to –3.8 EUR/MWh.
A better site (38.7 vs 35 %) $\approx$ –8 EUR/MWh.
OX2-type parameters (1,480 EUR/kW, 38.7 %):
 ~ 62 EUR/MWh.

Scale: where it helps

CAPEX/MW shows little scale effect in Finnish FIDs (277 MW: 1.44; 472 MW: 1.48 EUR m/MW). Scale helps via O&M and site choice. Hyperscalers prefer developers that deliver at scale, but the PPA market remains a buyer's market: scale buys access, not a premium.

CO-LOCATED BATTERY

The battery pays only if revenues stay near today's level

Standalone 25 MW / 50 MWh, NPV at 7 %

Case	kEUR/MW/yr	EUR m	IRR
Low	45	-6.5	-1.5 %
Base	65	-2.5	4.1 %
High = Jun 2026 level	89	+2.1	9.4 %
Break-even	77.8	0.0	7.0 %
Base, 20 % CAPEX saving	65	+0.4	7.5 %

A revenue decline of more than ~13 % from today's level makes the battery uneconomic.

Market is saturating

2-hour index Finland: 89 kEUR/MW/yr in Jun 2026, lower again in Jul as FCR and mFRR prices fell ~30 %. Installed ~1.5 GW; connection agreements point to ~4 GW.

Co-location helps the battery only

Each 10 % CAPEX saving from a shared grid connection lowers break-even revenue by ~7 kEUR/MW/yr. Wind cash flows are unchanged.

Source: model tab BESS (100 % equity, after tax, 15 years, CAPEX 287 EUR/kWh, Modo Energy 2026). Clean Horizon Storage Index Finland (Jun, Jul 2026, via ESS News); Solarplaza (Jul 2026); Fingrid half-year report H1 2026. O&M 1.5 % and optimiser fee 8 %: to verify. CAPEX saving share: own assumption.

14 checks against sources and hand calculations

Selected validation checks

Check	Result
Capture rate vs KYOS fundamental model	60.2 % vs 60 % base
LCOE hand calculation	47.8 vs 47.9 EUR/MWh
LCOE bridge to IRENA Finland 2025	38.9 EUR/MWh, gap explained
Break-even plausibility	Project IRR = WACC at NPV = 0
Wind data: Fingrid vs ENTSO-E	≈ 1 % from 2024; 2023 errors found
Recalculation and IRR cross-check	Identical in 18 scenario runs

How AI (Claude) was used

Formula audit, source search, Python pipeline drafts, scripted scenario runs. Every value entered by the author after reading the source; results re-checked by hand and recalculation.

Main limitations

Long-term prices from one MSc thesis · price and capture set independently · P50 only, single DSCR, no IDC or DSRA · fleet profile as park proxy

APPENDIX

Scenario grids: market price level × capture rate

100 % PPA, 50 EUR/MWh, DSCR 1.30x

Equity IRR	Capture Low	Base	High
Price Low	−5.4 %	−2.4 %	−0.7 %
Price Base	−2.2 %	−0.3 %	0.9 %
Price High	0.2 %	1.6 %	2.6 %

Full merchant, DSCR 1.60x

Equity IRR	Capture Low	Base	High
Price Low	−12.1 %	−7.0 %	−4.3 %
Price Base	−6.6 %	−3.4 %	−1.1 %
Price High	−2.5 %	0.1 %	2.1 %

Capture rate Low / Base / High: 50 / 60 / 70 %. Price paths: Low and High from Lampela (2023) storylines, Base from EEX futures (2028, 2030) and Lampela (2040, 2050). Low and High capture rates and High prices after 2030 are own assumptions.